Release notes: v0.1.0

Repository source: RELEASE_NOTES.md

Release date: September 6, 2026
Author: Bruce J. Swihart
Status: AI-assisted, non-peer-reviewed research draft seeking independent mathematical verification

Canonical repository: github.com/swihart/minvol-degree3-spectral-bound

Versioned release: v0.1.0

Main manuscript: PDF | LaTeX source

Purpose of this release

Version v0.1.0 is the initial public research draft of a proposed degree-3 spectral refinement of Nishioka's lower bound for the three-dimensional Blaschke-Lebesgue problem.

The proposed bound is

$$\mathrm{Vol}(K) \geq \frac{\pi}{1914}\left(259 - 27\sqrt{\frac{15183}{17303}}\right) d^3 \approx 0.3836027047 d^3.$$

The proposed contribution is an independent spectral mechanism: the proof separates the degree-3 curvature energy, derives a new determinant inequality for that component, and uses it to cap the portion of the total curvature budget that can receive Nishioka's degree-3 coefficient.

Included materials

This release contains:

- the integrated LaTeX paper and compiled PDF;
- expanded bridge lemmas and a detailed degree-3 tensor derivation;
- a claim-dependency graph, proof ledger, and internal proof audit;
- exact symbolic Python checks and exhaustive pairing enumeration;
- independent base-R pairing and numerical checks;
- automated GitHub Actions workflows for Python, R, document builds, best-effort PDF canonicalization, metadata checks, and PDF preflight;
- reproducibility, clean-clone, licensing, citation, and AI-provenance records; and
- typeset PDF counterparts of all Markdown documentation.

Verification status

The finite algebraic identities and constants are checked with exact symbolic arithmetic. Independent Python and R implementations agree numerically, and the repository has passed clean hosted builds and a recorded clean-clone reproduction.

The release-candidate smoke test ran every exact Python check, independent R check, paper build, Markdown-PDF build, metadata preflight, and PDF validity and checksum check from a fresh clone. All substantive stages passed. Two secondary rendered documentation PDFs differed from their committed copies by two bytes; no source or mathematical output changed. Exact PDF byte identity is therefore reported as a build-detail diagnostic rather than used as a release gate.

The exact tagged commit must pass the three hosted GitHub Actions jobs on `main`, and the tag-triggered workflow must also pass before the GitHub release is published. These checks establish reproducibility of the supplied computations and successful document generation. They do not constitute independent subject-matter review or peer review of the complete proof.

Scope and limitations

HYRA publicly claims substantially stronger analytic and computer-assisted bounds by a different method. This release does not claim the strongest currently proposed numerical coefficient. It also does not prove the Meissner conjecture or identify a minimizing body.

Errors, missing hypotheses, normalization issues, and literature references may still be found. Corrections should preserve the v0.1.0 record and be issued under a new version tag.

Preferred citation

Bruce J. Swihart, *A Degree-Three Spectral Refinement of Nishioka's Lower Bound for the Three-Dimensional Blaschke-Lebesgue Problem*, version v0.1.0, public research draft, September 6, 2026.

The machine-readable citation is in `CITATION.cff`.